DSA

Practical 6

Circular Queue

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Branch: CSE (AIML)

Batch: A2

CODE:-

```
#include <iostream>
using namespace std;

struct Node{
   int data;
   Node* next;
};

struct Node* front = NULL;
struct Node* rear = NULL;

void enqueue(int val){
   struct Node* ptr;
   ptr = new Node();
   ptr->data = val;
   ptr->next = NULL;
   if(front == NULL){
      front = rear = ptr;
      rear->next = front;
   } else {
      rear->next = ptr;
      rear = ptr;
      rear->next = front;
   }
   cout << val << " is inserted." << endl;
}

void dequeue(){
   if(front == NULL){
      cout << "Queue is empty." << endl;
```

```
        return;
    }

    struct Node* temp = front;
    if(front == rear){
        cout << front->data << " is deleated." << endl;
        delete front;
        front = rear = NULL;
    }
    else{
        cout << front->data << " is deleated." << endl;
        front = front->next;
        rear->next = front;
        delete temp;
    }
}

void display(){
    if(front == NULL){
        cout << "Queue is empty." << endl;
        return;
    }
    struct Node* temp = front;
    while(true){
        cout << temp->data << " ";
        temp = temp->next;
        if(temp == front) break;
    }

    cout << endl;
}
```

```
void peek(){
    if(front == NULL){
        cout << "Queue is empty." << endl;
    } else {
        cout << "Front element: " << front->data << endl;
    }
}

int main(){
    int choice;
    do {
        cout << "\n--- Menu ---\n";
        cout << "1. Enqueue\n";
        cout << "2. Dequeue\n";
        cout << "3. Display\n";
        cout << "4. Peek\n";
        cout << "5. Exit\n";
        cout << "Enter choice: ";

        if(!(cin >> choice)){
            cin.clear();
            cin.ignore(10000, '\n');
            cout << "Invalid input.\n";
            continue;
        }

        switch(choice) {
            case 1: {
                int val;
                cout << "Enter value to enqueue: ";
```

```
            while(!(cin >> val)){
                cin.clear();
                cin.ignore(10000, '\n');
                cout << "Invalid input. Enter an integer: ";
            }
            enqueue(val);
            break;
        }
        case 2:
            dequeue();
            break;
        case 3:
            display();
            break;
        case 4:
            peek();
            break;
        case 5:
            cout << "Exiting.\n";
            break;
        default:
            cout << "Invalid choice.\n";
    }
  } while(choice != 5);

  return 0;
}
```

OUTPUT:-

/*

--- Menu ---

1. Enqueue

2. Dequeue

3. Display

4. Peek

5. Exit

Enter choice: 1

Enter value to enqueue: 45

45 is inserted.

--- Menu ---

1. Enqueue

2. Dequeue

3. Display

4. Peek

5. Exit

Enter choice: 1

Enter value to enqueue: 85

85 is inserted.

--- Menu ---

1. Enqueue

2. Dequeue

3. Display

4. Peek

5. Exit

Enter choice: 3

45 85

--- Menu ---

1. Enqueue

2. Dequeue

3. Display

4. Peek

5. Exit

Enter choice: 4

Front element: 45

--- Menu ---

1. Enqueue

2. Dequeue

3. Display

4. Peek

5. Exit

Enter choice: 2

45 is deleated.

--- Menu ---

1. Enqueue

2. Dequeue

3. Display

4. Peek

5. Exit

Enter choice: 5

Exiting.

*/